

Interpreting Bones and Shell



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To make interpretations about the populations of animals represented in the archaeological assemblage, zooarchaeologists calculate secondary data, which are estimates about the age, sex, and body size of individuals, and also the population-level structure of these characteristics. Through the archaeological assemblage, it is also possible to look at human strategies for hunting, butchering, cooking, and manipulation of the herd structure. All of these are interpretations that then allow zooarchaeologists to answer research questions about how humans interacted with animal populations.

Secondary data are data extrapolated from attributes of the skeletal elements examined, especially at the level of the population contained within the archaeological assemblage.

Body size, for example, can be estimated based on the length or width of specific features on specific bones. By creating a simple histogram recording the counts of bones of specific size ranges in a taxon, it is possible to identify whether those animals belong to a single-size population or multiple populations of different sizes, such as dimorphic males and females, animals of different age classes (i.e., juveniles versus adults), or animals using different hunting strategies (e.g., trapping fish with a net versus hook-and-line fishing). Alternatively, we can look at age at death to compare the population structures of domesticated animals being managed for different resources, such as sheep being killed young for meat or being kept to old ages to maximize wool production.

Looking at which bones are present and which are missing in an assemblage is also instructive about butchering patterns, where animals were killed and butchered, and what meat was preferred for cooking. For example, a site with only the lower limbs, neck, and heads of deer is what we expect to see at a hunting camp where deer were butchered, as only the parts of the body with large amounts of meat (i.e., upper legs and torso) were brought back home (somewhere else), where the meat-bearing bones were eventually deposited. We can also look at butchery practices by recording patterns of cut marks on specific bones, which can indicate the ways that animal carcasses were divided. Cooking patterns are also evident from which bones we find evidence of burning on, and where on those bones. This can indicate whether certain cuts of meat were roasted or boiled or stewed in liquid.